HIV AIDS AND STIS Notes-1

Health records and information technology (Kenya Medical Training College)

Scan to open on Studocu

Studocu is not sponsored or endorsed by any college or university
Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

HIV, AIDS AND STIS

MARCH 2021 BLOCK II

TOPIC 1: FUNDAMENTALS OF HIV

DEFINATION OF TERMS

HIV-Stands for human immuno-deficiency virus.

H=Human(refer to us)

I=Immunodeficiency (lack of protection from getting sick)

V=Virus (a type of germ in the body that can be cured). This is the virus that causes AIDS

AIDS-Stands for acquired immune deficiency syndrome.

A=Acquired(to get something)

I=Immune(the way the body fights disease)

D=Deficiency(not enough of something-in this case lack of protection from getting sick)

S=Syndrome (a group of symptoms or illness)

AIDS: This is state when a person gets a group of serious illness and opportunistic infection that develop after speedy multiplication of HIV virus in the body with consequent reduction of CD4.This results into the body's inability to fight other invading germs thus making the person very prone to diseases that take advantage of the lowered immunity.

HIV infected; is when HIV virus has entered a person's body. A person who is HIV-infected might be very healthy and not have any signs of illness for a very long time but they can pass the virus to others.

STIs stands for sexually transmitted infections/ diseases. WHO define STI as communicable disease mainly transmitted through sexual intercourse with an infected person. They are caused by a number of micro-organisms eg virus,bacteria etc

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

How does a person know they have HIV or AIDS?

HIV infection can be diagnosed with a simple blood test.This is usually done in VCT(Voluntary counselling and testing) clinics or through PITC(Patient initiated testing and counselling)

HIV test is conducted to pregnant mothers as a routine part of antenatal care.

AIDS is diagnosed by clinical and lab tests, usually with blood samples, done at the hospital.

ORIGIN & HISTORY OF HIV/AIDS:

ORIGIN

When and where the HIV virus first emerged is probably going to remain a mystery for many years to come. While several theories have been put forward, there is no conclusive single agreement on the origin of HIV/AIDS. Some of the mostly acknowledge theories about the origin of HIV include the following

• Mysterious origins e.g the tail of the comet theory

• Religious theories (God's wrath and witchcraft)

• Monkey origin theories (with four theories)

• Conspiracy theories

• The calculation theory

1. **Mysterious origin theory**

Tries to account for the seemingly mysterious origin of HIV by locating it out of this world. Suggest that viral materials was carried in the tail gases of a comet passing close to the earth and its material was deposited, subsequently infecting nearby sate in people. Although one or two famous astronomers have been linked to this theory, in the popular press, these scientists deny the possibility of this extraterrestrial phenomena and any personal connection to the theory.1

2. **Religious theories (God's wrath and witch craft)**

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Certain segments of the population have openly stated that AIDS is God's wrath since the scripture condemn the homosexual practice in which AIDS was 1st observed in western world Rather than its being considered a visitation from God; many Africans believe that AIDS is any caused by another supernatural power-witchcraft and they use anti-witchcraft rituals and objects to counteract the infection.

3. **Monkey origin theories**

HIV is a lentivirus called simian immunodeficiency virus (SIV). The research done claimed that chimpanzees were the source of HIV 1, and that the virus had at some point crossed species from chimps to humans. How HIV crossed species is explained by theories such as

-The oral polio vaccine (OPV) Theory

-The hunter's theory

-The contaminated needle theory

-Colonization theory.

4. **Conspiracy theories**

Americans believed HIV was manufactured as part of a biological warfare programme, designed to wipe out large numbers of black and homosexual people

5. **The calculation theory**

The latest in the origin of HIV. Opponents of SIV-human transmission remain unimpressed by the evidence in the monkey theories and argue that viral sequencing of HIV strains indicate that HIV has been around probably for hundreds of years

In 2000 when a team of scientist using computer technology to study the structure of HIV calculated the rate at which the virus mutates for the HIV viral sub-types to have a common ancestors. This process revealed HIV originated around 1930 in rural areas of central Africa, where the virus may have been present for many years in isolated communities

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

The virus probably did not spread because members of these rural communities had limited contact with people from other areas. But in 1960s and 1970s, political upheaval, wars, drought, and famine forced many people from these rural areas to migrate to cities to find jobs. During this time, the incidence of sexually transmitted infections, including HIV infection, accelerated and quickly spread throughout Africa.

What caused the rapid spread of the HIV epidemic in the latter half of the 20th century: There are a number of factors that may have contributed to the sudden global spread of HIV in the latter half of the 20th century and they include

a. International travel,

b. The blood industry

c. Widespread drug use

HISTORY OF HIV

History in USA:

HIV virus began to emerge in the United States in the 1980s, the HIV epidemic has frequently been linked to gay, bisexual and men who have sex with men(MSM).The 1st official report on the virus was published by center for disease control(CDC) in 1981 and detailed a case of 5 young gay men who were hospitalized with serious infections later the New York times reported 41 homosexual had been diagnosed with Kaposi sarcoma and 8 had died less than 24 months after dx made.

HIV history in Kenya

In Kenya, the first HIV case was recorded in 1984.

It began between 1983 and 1985 when 26 cases of Aids were reported in Kenya. Sex workers were the first group affected. A study from 1985 reported a HIV prevalence of 59% amongst a group of sex workers in Nairobi.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Epidemiology of HIV

HIV continues to be a major global public health issue, having claimed almost 33 million lives so far. However, with increasing access to effective HIV prevention, diagnosis, treatment and care, including for opportunistic infections, HIV infection has become a manageable chronic health condition, enabling people living with HIV to lead long and healthy lives.

There were an estimated 38.0 million people living with HIV at the end of 2019.

As a result of concerted international efforts to respond to HIV, coverage of services has been steadily increasing. In 2019, 68% of adults and 53% of children living with HIV globally were receiving lifelong antiretroviral therapy (ART).

A great majority (85%) of pregnant and breastfeeding women living with HIV also received ART, which not only protects their health, but also ensures prevention of HIV transmission to their newborns.

At the end of 2019, an estimated 81% of people living with HIV knew their status. 67% were receiving antiretroviral therapy (ART) and 59% had achieved suppression of the HIV virus with no risk of infecting others; about 27 million adolescent boys and men in east and southern Africa had received prevention services including VMMC.

By June 2020, 26 million people were accessing antiretroviral therapy, marking a 2.4% increase from an estimate of 25.4 million at the end of 2019. By comparison, treatment coverage increased by an estimated 4.8% between January and June of 2019.

The number of new people starting treatment is far below expectation due to the reduction in HIV-testing and treatment initiation and ARV disruptions that occurred during the COVID-19 pandemic. By end 2020, testing and treatment rates showed steady but variable recovery.

Nevertheless, between 2000 and 2019, new HIV infections fell by 39% and HIV-related deaths fell by 51%, with 15.3 million lives saved due to ART. This achievement was the result of great

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

efforts by national HIV programmes supported by civil society and international development partners.

But success has been variable by region, country and population. However, not everyone is able to access HIV testing, treatment and care. Even prior to the COVID-19 pandemic, reduction of new infections and deaths had plateaued; global 90/90/90 targets for 2020 are at risk of being missed unless rapid action is taken.

Due to gaps in HIV services, 690 000 people died from HIV-related causes in 2019 and 1.7 million people were newly infected.

To reach the new proposed **global 95/95/95 targets**, (95% of the total population should be tested, 95% of those diagnosed positive should receive treatment and 95% of those receiving treatment should achieve viral suppression), we will need to redouble our efforts to avoid the worst-case scenario of a half million excess deaths in Sub Saharan Africa, increasing HIV infections due to HIV service disruptions during COVID-19, and the slowing public health response to HIV.

Interventions will need to focus on the populations left-behind: Key population groups and their sexual partners accounted for over 62% of all new HIV infections globally among the age group 15-49 years in 2019. In eastern European and central Asia, Asia and the Pacific, western and central Europe, and North America, and the Middle East and North Africa, these groups accounted for over 95% of new HIV infections in each of these regions.

WHO defines **key populations as people in populations who are at increased HIV risk in all countries and regions. Key populations include: men who have sex with men; people who inject drugs; people in prisons and other closed settings; sex workers and their clients; and transgender people**.

Increased HIV vulnerability is often associated with legal and social factors, which increases exposure to risk situations and creates barriers to accessing effective, quality and affordable HIV prevention, testing and treatment services. Prioritizing key populations in the HIV response

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

with appropriate interventions would have the biggest impact on the epidemic and reduce new infections.

In addition, given their life circumstances, a range of other populations may be particularly vulnerable, and at increased risk of HIV infection, such as adolescent girls and young women in southern and eastern Africa and indigenous peoples in some communities.

Over two thirds of all people living with HIV live in the WHO African Region (25.7 million). While HIV is prevalent among the general population in this region, an increasing number of new infections occur among key population groups.

HIV/AIDS IN KENYA:

Kenya has an average HIV prevalence rate of 6% and with about 1.6m people living with HIV infection, it is one of the six HIV high burden countries in Africa. The western part of the country through Homabay,Siaya and Kisumu are the most affected with HIV with rates of 25.7%, 23.7% and 19.3% respectively

In Kenya women were more likely to be infected(8.4%) than men (5.4%).In particular young women aged 15-24 years are 4 times more likely to be infected than young men in the same age group(5.6% vs 1.4%). An estimated 6.7% adult in rural areas are infected with HIV compared to estimated 8.4% in urban areas

Uncircumcised men are 3 times more likely to be infected with HIV than circumcised men(13.2% vs 3.9%)respectively. 80-90% of infections are in the 15-49 years age group, while 5-10% occurs in children less than 5 years old.

The Biology of HIV

Although many viruses cause suppression of the immune system, only one wipes it out completely, and that is the previously mentioned. It is worth discussing the biology of HIV virus,

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

which can lead to the well-known AIDS, so that its full effects on the immune system can be understood.

The overview of the immune system

Our immune system is essential for our survival. Without an immune system, our bodies would be open to attack from bacteria, viruses, parasites, and more. It is our immune system that keeps us healthy as we drift through a sea of pathogens.

The immune system protects the body against disease or other potentially damaging foreign bodies. When functioning properly, the immune system identifies and attacks a variety of threats, including viruses, bacteria and parasites, while distinguishing them from the body's own healthy tissue. It is a vast network of cells and tissues that are constantly on the lookout for invaders, and once an enemy is spotted, a complex attack is mounted.

The immune system is spread throughout the body and involves many types of cells, organs, proteins, and tissues. Crucially, it can distinguish our tissue from foreign tissue — self from non-self. Dead and faulty cells are recognized and cleared away by the immune system.

The body's immune system is mainly the lymphatic system which consists of a collection of barriers, cells, and soluble proteins that interact and communicate with each other in extraordinarily complex ways as indicated in the fig **2.1 below**;

These organs and tissue of the immune system includes; **the bone marrow, spleen, thymus and lymph nodes**. The spleen is the largest lymphatic organ in the body contains white blood cells that fight infection or disease. **The thymus** is where **T-cells mature.** T-cells help destroy infected or cancerous cells. Lymph nodes produce and store cells that fight infection and disease. Lymphocytes and leukocytes are small white blood cells that play a large role in defending the body against disease.The two types of lymphocytes are **B-cells and T-cells.**Leukocytes are white blood cells that identify and eliminate pathogens.

- **Thymus** — a gland between the lungs and just below the neck.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- **Spleen** — an organ that filters the blood. It sits in the upper left of the abdomen.
- **Bone marrow** — found in the center of the bones, it also produces red blood cells.
- **Lymph nodes** —small glands positioned throughout the body, linked by lymphatic vessels.

The modern model of immune function is organized into **three phases** based on the timing of their effects. The three temporal phases consist of the following:

i. **Barrier defenses** such as the skin and mucous membranes, which act instantaneously to prevent pathogenic invasion into the body tissues

ii. The rapid but nonspecific **innate immune response**, which consists of a variety of specialized cells and soluble factors

iii. The slower but more specific and effective **adaptive immune response**, which involves many cell types and soluble factors, but is primarily controlled by white blood cells (leukocytes) known as **lymphocytes**, which help control immune responses. These cells can be divided into three classes based on function:

a. **Phagocytic cells**, which ingest pathogens to destroy them

b. **Lymphocytes**, which specifically coordinate the activities of adaptive immunity

c. **Cells containing cytoplasmic granules**, which help mediate immune responses against parasites and intracellular pathogens such as viruses.

All the cells of the immune response as well as of the blood arise by differentiation from **hematopoietic stem cells.**Lymphocytes are the primary cells of adaptive immune responses its two basic types are; **B Cells, T Cells.** There are two other types of lymphocytes namely**; Plasma Cells,** and **Natural Killer Cells.**

B and T cellsare identical morphologically with a large central nucleus surrounded by a thin layer of cytoplasm and they are both initially developed from bone marrow. They are distinguished from each other by;

✓ Their surface protein markers

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- ✓ The molecules they secrete.
- ✓ **B cells** mature in **red bone marrow** while **T cells** mature in the **thymus**.

N/B; T-cells are produced from the bone marrow and they migrate from to the thymus gland where they further mature.

B cells and T cells are found in many parts of the body, circulating in the bloodstream and lymph, and residing in secondary lymphoid organs, including the spleen and lymph nodes. The human body contains approximately **1012 lymphocytes**.

i. B Cells

B cells are immune cells that function primarily by producing antibodies. An **antibody** is any of the group of proteins that binds specifically to pathogen-associated molecules known as antigens. An **antigen** is a chemical structure on the surface of a pathogen that binds to T or B lymphocyte antigen receptors. Once activated by binding to antigen, B cells differentiate into cells that secrete a soluble form of their surface antibodies. These activated B cells are known as plasma cells.

ii. T Cells

The **T cell**, on the other hand, does not secrete antibody but performs a variety of functions in the adaptive immune response. Different T cell types have the ability to either secrete soluble factors that communicate with other cells of the adaptive immune response or destroy cells infected with intracellular pathogens.

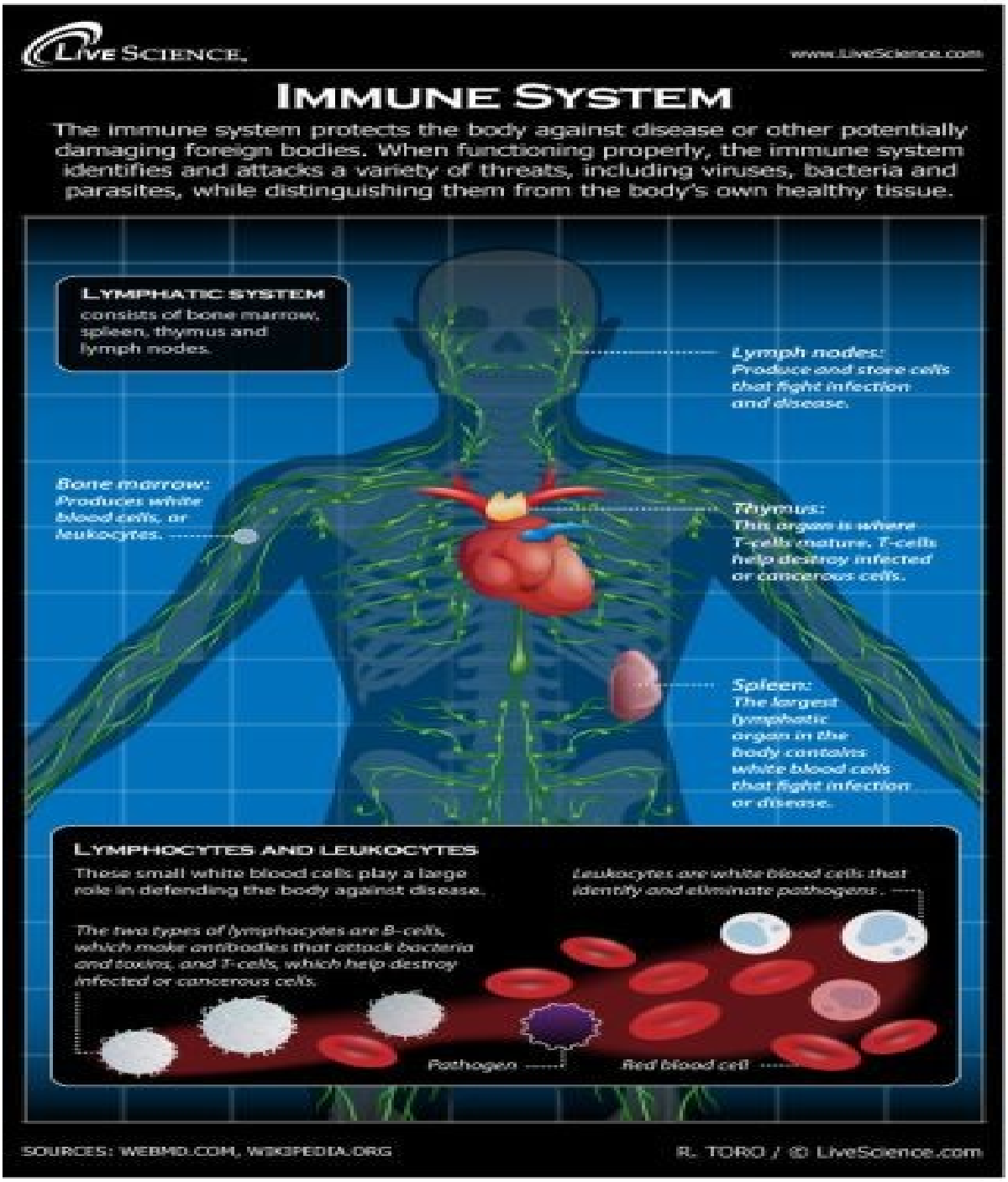
iii. Plasma Cells

Another type of lymphocyte of importance is the plasma cell. A **plasma cell** is a B cell that has differentiated in response to antigen binding, and has thereby gained the ability to secrete soluble antibodies. These cells differ in morphology from standard B and T cells in that they contain a large amount of cytoplasm packed with the protein-synthesizing machinery known as rough endoplasmic reticulum.

iv. Natural Killer Cells

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

A fourth important lymphocyte is the natural killer cell, a participant in the innate immune response. A **natural killer cell (NK)** is a circulating blood cell that contains cytotoxic (cell-killing) granules in its extensive cytoplasm. It shares thismechanism with the cytotoxic T cells of the adaptive immune response. NK cells are among the body's first lines of defenseagainst viruses and certain types of cancer.



This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Fig 2.1 The human immune system

The immune system protects its host from infection by use of two categories of defenses namely;

1. **<u>Non-Specific Defenses</u>**

   i. First Line of Defense

      a. Skin & mucous membranes

      b. Lysozyme (enzyme that breaks down cell walls of bacteria; contained in sweat, tears, mucus, saliva)

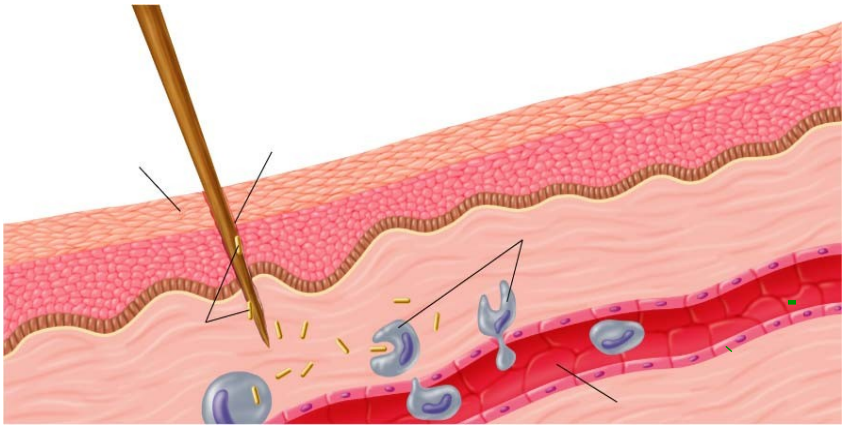


Fig the skin as a first line of defense

   ii. Second Line of Defense

   ✓ Inflammatory response

   ✓ Phagocytes (WBCs)

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

2. **Specific Defenses: The Immune Response**

i. **Lymphocytes (WBCs)**

a. B cells

b. T cells

ii. **Humoral Immunity**

a. Antibodies (proteins) produced by B cells attack the pathogen

iii. **Cell-Mediated Immunity**

- Helper T cells (aka CD4 cells)
- Cytotoxic (Killer) T cells

In many species, there are two major subsystems of the immune system:

- **Innate immune system**
- **Adaptive immune system**.

Both subsystems use **humoral immunity** and **cell-mediated immunity** to perform their functions.

In humans, the blood–brain barrier, blood–cerebrospinal fluid barrier, and fluid–brain barriers separate the peripheral immune system from the neuro-immune system, which protects the brain.

These physical barriers prevent pathogens such as bacteria and viruses from entering the organism. If a pathogen breaches these barriers, the **innate immune system** provides an immediate, but non-specific response. If pathogens successfully evade the innate response, vertebrates possess a second layer of protection, the **adaptive immune system**, which is activated by the innate response. Here, the immune system adapts its response during an infection to improve its recognition of the pathogen. This improved response is then retained after the pathogen has been eliminated, in the form of an **immunological memory**, and allows the adaptive immune system to mount faster and stronger attacks each time this pathogen is

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

encountered. If the immune system encounters a pathogen, for instance, a bacterium, virus, or parasite, it mounts a so-called immune response.

How an immune response works

The immune system needs to be able to tell self from non-self. It does this by detecting proteins that are found on the surface of all cells. It learns to ignore its own or self-proteins at an early stage.

An antigen is a molecule on a pathogen that triggers an immune response that are recognized as "non-self". In many cases, an antigen is a bacterium, fungus, virus, toxin, or foreign body. But it can also be one of our own cells that is faulty or dead. Initially, a range of cell types works together to recognize the antigen as an invader.

The role of B lymphocytes

Once B lymphocytes spot the antigen, they begin to secrete antibodies (antigen is short for "**antibody generators**"). **Antibodies** are special proteins that lock on to specific antigens that are produced in the in the humoral immune response. Each B cell makes one specific antibody. For instance, one might make an antibody against the bacteria that cause pneumonia, and another might recognize the common cold virus. Antibodies are part of a large family of chemicals called **immunoglobulins**, which play many roles in the immune response. They include:

i. **Immunoglobulin G (IgG)** — marks microbes so other cells can recognize and deal with them.

ii. **IgM** — is expert at killing bacteria.

iii. **IgA** — congregates in fluids, such as tears and saliva, where it protects gateways into the body.

iv. **IgE** — protects against parasites and is also to blame for allergies.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

v. **IgD** — stays bound to B lymphocytes, helping them to start the immune response.

N/B; Antibodies lock onto the antigen, but they do not kill it, only mark it for death. The killing is the job of other cells, such as phagocytes.

The role of T lymphocytes

There are distinct types of T lymphocytes:

- ✓ **Helper T cells (Th cells)** — they coordinate the immune response. Some communicate with other cells, and some stimulate B cells to produce more antibodies. Others attract more T cells or cell-eating phagocytes.
- ✓ **Killer T cells (cytotoxic T lymphocytes)** — as the name suggests, these T cells attack other cells. They are particularly useful for fighting viruses. They work by recognizing small parts of the virus on the outside of infected cells and destroy the infected cells.

Everyone's immune system is different but, as a general rule, it becomes stronger during adulthood as, by this time, we have been exposed to more pathogens and developed more immunity. That is why teens and adults tend to get sick less often than children.

Once an antibody has been produced, a copy remains in the body so that if the same antigen appears again, it can be dealt with more quickly. That is why with some diseases, such as chickenpox, you only get it once as the body has a chickenpox antibody stored, ready and waiting to destroy it next time it arrives. This is called immunity. There are three types of immunity in humans called innate, adaptive, and passive:

i. Innate immunity

We are all born with some level of immunity to invaders. Human immune systems, similarly to those of many animals, will attack foreign invaders from day one. This innate immunity includes the external barriers of our body — the first line of defense against pathogens — such as the skin and mucous membranes of the throat and gut. This response is more general and non-specific. If the pathogen manages to dodge the innate immune system, adaptive or acquired immunity kicks in.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

ii. Adaptive (acquired) immunity

This defense from pathogens develops as we go through life. As we are exposed to diseases or get vaccinated, we build up a library of antibodies to different pathogens. This is sometimes referred to as **immunological memory** because our immune system remembers previous enemies.

iii. Passive immunity

This is a type of immunity that is "**borrowed"from another source,** but it does not last indefinitely. For instance, a baby receives antibodies from the mother through the placenta before birth and in breast milk following birth. This passive immunity protects the baby from some infections during the early years of their life.

iv. Immunizations

Immunization introduces antigens or weakened pathogens to a person in such a way that the individual does not become sick but still produces antibodies. Because the body saves copies of the antibodies, it is protected if the threat should reappear later in life.

A primary infection is like a **race** between the **pathogen** and the **immune system.** The pathogen bypasses barrier defenses and starts multiplying in the host's body. During the first 4 to 5 days, the **innate immune** response will partially control, but not stop, pathogen growth.

As the **adaptive immune response** gears up, however, it will begin to clear the pathogen from the body, while at the same time becoming stronger and stronger. When following antibody responses in patients with a particular disease such as a virus, this clearance is referred to as **seroconversion** (sero- = "serum").

Seroconversionis the reciprocal relationship between virus levels in the blood and antibody levels. As the antibody levels rise, the virus levels decline, and this is a sign that the immune response is being at least partially effective (partially, because in many diseases, seroconversion does not necessarily mean a patient is getting well).

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

An excellent example of this is sero-conversion during HIV disease (see **fig 2.2 below**). Notice that antibodies are made early in this disease, and the increase in anti-HIV antibodies correlates with a decrease in detectable virus in the blood. Although these antibodies are an important marker for diagnosing the disease, they are not sufficient to completely clear the virus.

Several years later, the vast majority of these individuals, if untreated, will lose their entire adaptive immune response, including the ability to make antibodies, during the final stages of AIDS.

Seroconversion show the rise of anti-HIV antibody levels and the associated decline in measurable virus levels which happens during the first several months of HIV disease. Unfortunately, this antibody response is ineffective at controlling the disease, as seen by the progression of the disease towards AIDS, in which all adaptive immune responses are compromised.

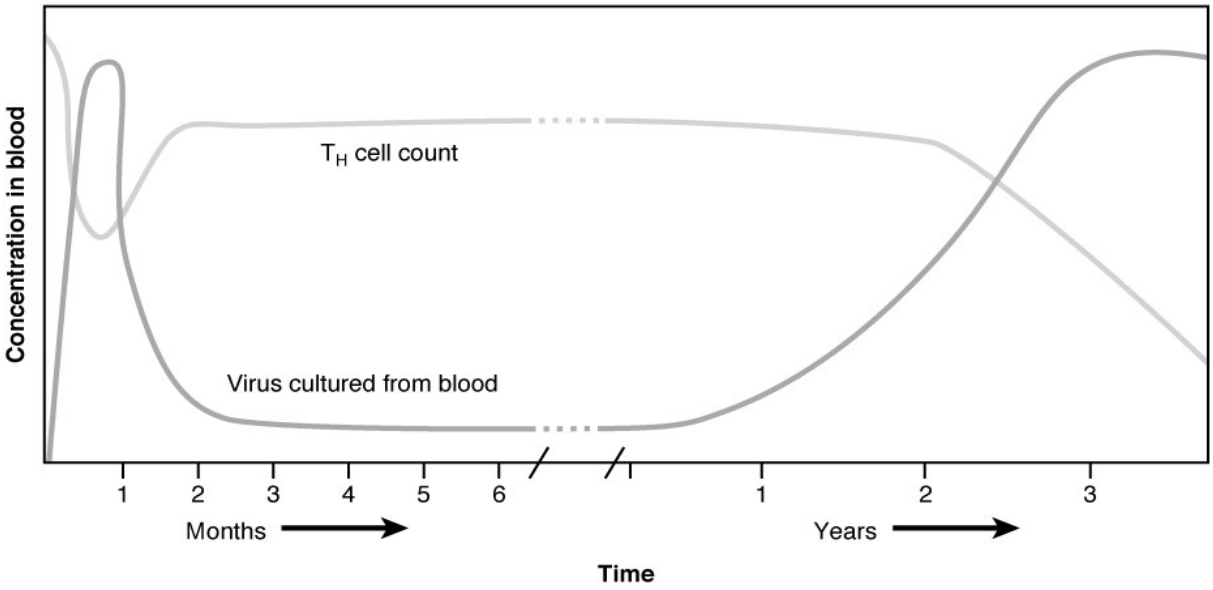


Fig 2.2 HIV disease progression

Natural immunity to HIV and AIDS;

There are individuals with HIV immunity which have really challenged scientists for over a decade. The question is how is it that the immune systems of some seem impervious to a virus that kills 2 million people around the globe each year?

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Researchers have focused on a few proteins that may hold the key to this puzzle as well as offer the potential for new HIV treatments. These are;

i. CCR5,
ii. Human leukocyte antigen (HLA)
iii. Cystatin
iv. CD4

i. **CCR5**; A new study at the University of Southern California showed that mice with a mutation in the gene that encodes **CCR5** have immunity to HIV. Their work provides "proof of concept for a new approach to HIV treatment." CCR5 is a protein found on the surface of human immune-system cells. Essentially, they work as a lock that HIV opens in order to enter the cells. In this study, researchers took mice already infected with HIV and injected them with stem cells containing a specific mutation in the CCR5 gene. They found the injected cells were able to fight and destroy HIV, and the mice were able to fight off other infections, too. Because stem cells reproduce indefinitely, these mutant stem cells could provide a permanent supply of HIV-resistant immune cells, according to the researchers. This procedure that was developed in collaboration with scientists at biotech company Sangamo BioSciences is currently being tested in humans in Phase 1 clinical trials. It was inspired by a 2009 New England Journal of Medicine case report that described a patient with both HIV and leukemia. After undergoing a bone marrow transplant from a donor with the CCR5 mutation — known as the CCR5-delta 32 mutation — the patient became HIV-free and no longer required anti-AIDS drugs.

This mutation in CCR5 is therefore associated with natural immunity to HIV in about 10 percent of Caucasian people. Scientists suspect that its commonness comes from being spared by deadly plagues in the distant past. However, there is disagreement over which disease or diseases influenced the mutation over time. Much research has shown that the mutation may have given some people immunity to the waves of bubonic plague that swept through Europe during the 12th through 15th centuries. But University of Berkeley researchers suggest

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

smallpox is a likely cause for the mutation's spread. In a 2003 report in the Proceedings of the National Academies of Science, the scientists explained that smallpox was around far longer than the plague and killed far more people. And smallpox especially affected younger children, who were not yet old enough to reproduce.

In a 2006 study, Johns Hopkins University researchers found that the mutation reduced infection by the hepatitis B virus, as well. They concluded that "a diverse group of infectious disease, rather than a single, deadly pathogen," may have been the driving force behind the mutation's prevalence.

N/B: Because the CCR5 mutation does not provide HIV immunity in all populations, researchers have looked at other proteins that may bestow a natural advantage in fighting off the virus. These include;

ii. **Cystatin;** this is a protein that was found by researchers in 2008 at the University of Manitoba, studied Kenyan women who were still HIV-free after working as prostitutes for at least three years. The scientists found increased levels of cystatin, which is known to interfere with the ability of HIV to reproduce.

iii. **Human leukocyte antigen (HLA);** Studies of Zambians have found the influence of the protein or human leukocyte antigen (HLA). It is also called "**elite controllers**" – these are people whose cells are able to effectively attack and destroy HIV due to these HLA. They may never experience symptoms even though they are infected with the virus.

iv. **CD4**.; this is another protein that as with CCR5, HIV must interact with it in order to enter person's immune cells, and some say it may make a better drug target than CCR5.

Recently Peter Kwong, a scientist at the National Institute of Allergy and Infectious Diseases, led a team that investigated a protein produced by people immune to HIV that binds to HIV and to CD4. The researchers concluded that fully understanding how this protein binds to both the virus and the human cells could lead to the creation of an HIV vaccine. "The CCR5 binding site is only revealed to the virus after it binds to CD4," Kwong said. "So although CCR5 is an extremely

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

good drug target, the CD4 site is much better because it must always be accessible for HIV to get into the cell."

TRANSMISSION OF HIV:

HIV can be transmitted in the following body fluids;

- Semen
- Vaginal fluids
- Blood
- Birth fluids
- Breast milk

HIV is not transmitted in these fluids unless there is also blood present;

- Urine
- Faeces
- Saliva
- Sweat
- Mucus

HIV/AIDS is transmitted through the following ways:

- Sexual contact with a person infected with the virus(unprotected sexual intercourse with infected person) NB: sexual transmission accounts for 87% of HIV transmission worldwide.
- From infected mother to her unborn child in the womb, during birth and after birth. It accounts for 30%
- Transfusion with infected blood or blood products accounts for 15%
- Cutting and skin piercing instruments e.g needles, razors, knives(e.g during circumcision)
- Occupational exposure like for medical personnel
- Deep kissing if one has open wounds in the mouth

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Ways HIV is not transmitted:

- Sharing food or a drinking cup
- Hugging
- Shaking hands
- Coughing or sneezing
- Being near a person with aids
- Sharing a latrine or toilet
- Mosquito or insect bite

HIV Types and Subtypes

One of the obstacles to treatment of the human immunodeficiency virus is its high genetic variability.

HIV can be divided into two major types,HIV type 1(HIV-1) and HIV type 2(HIV-2)

HIV-1 is related to viruses found in chimpanzees and gorillas living in western Africa.

HIV-2 viruses are related to viruses found in the endangered West Africa primate sooty mangabey.

HIV-1 viruses may be further divided into groups. Hiv-1 group M viruses predominate and are responsible for the aids pandemic.Group M can be further subdivided into subtypes based on genetic sequence data.Some of the subtypes are known to be more virulent or are resistance to different medications.

Likewise HIV-2 viruses are thought to be less virulent and transmissible than HIV-1 M group viruses although HIV-2 is known to cause HIV.

AIDS VIRUS LIFE CYCLE

HIV enter cell through interaction of an envelope glycoprotein (gp120) and the cellular receptor, CD4, and other co-receptor such as CCR5 and CXCR4.These receptors are expressedon

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

T-helper lymphocytes(CD4 lymphocytes), macrophages,dendritic cells(in lymph nodes and mucosal surfaces) and microglial cells in the brain.

Viruses that utilize the CCR5 co-receptors to gain entry into cells are referred to as R5 tropic, whereas those that use CXCR4 co-receptor are X4 tropic.

Most primary HIV infection are R5 tropic. The X4 tropic virus appear late in the course of HIV infection.The virus life cycle can be divided into a number of distinct steps, each of which is a current or potential therapeutic drug target.

Stages of replication of the HIV virus

1. Binding

Is also called attachment. During this step, viralgp 120 binds to the CD4 receptor and CCR5 or CXCR4 co-receptor on host cell surface. Post attachment inhibitor can be used to slow down this process.

2. Fusion

The HIV envelope and the CD4 cells fuse hence allowing the HIV virus to enter the CD4 cell. Fusion inhibitor acts as this process.

3. Reverse transcription

Inside the CD4 cellViral RNA is reverse transcribed into viral DNA .This process is mediated by the enzymes reverse transcriptase (RT).This allows the HIV to enter the CD4 cell nucleus and combine with the cells genetic material. This enzymes has no proof-reading function, and therefore the process is error prone and responsible for rapid development of drugs resistant mutation.

Inhibitors involved at this stage include non-nucleoside reverse transcriptase inhibitors and nucleoside reverse transcriptase inhibitors

4. Integration

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Inside the CD4 nucleus, HIV releases integrase enzyme and uses it to integrate its viral DNA into the DNA of the host cell.This is the step that establishes replication competent virus in the body and makes HIV incurable even with effective ART. Intergrase inhibitor can slow down this process.

5. **Replication**

Once integrated into the CD4 cell DNA, HIV begins to use the machinery of the CD4 cell to make long chains of HIV proteins which produce more HIV viruses.

6. **Assembly**

New HIV proteins and HIV RNA move to the surface of the cell and assemble into immature non-infectious HIV.

7. **Budding**

Newly formed immature HIV pushes itself out of the host CD4 cells and release the protease enzyme. This enzyme breaks up the long protein chains in the immature virus, creating mature infectious viruses which are then released to the bloodstream.

The protease inhibitor slows down this process.

Emerging issues in HIV infection

1. Gender based violence and

Gender-based violence has become common place in almost all societies. Violence has many faces. Within the household this can include **battering by an intimate partner, marital rape, and sexual abuse.**

Violence outside the home can include rape, sexual abuse, sexual harassment and assault. Various social, cultural, and religious norms produce and reinforce gender inequality and the stereotypical gender roles that underpin gender-based violence. Gender-based violence is a key factor in increasing risk of contracting HIV.

Violence increases vulnerability to HIV infection in several ways. Sexual violence can result in ‘direct transmission’ of HIV which can be the result of forced or coercive sexual intercourse with

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

an HIV infected partner. The biological risk of transmission in a violent sexual encounter is determined by the type of sexual exposure

2. Intimate partner violence and HIV

Intimate partner violence (IPV) is domestic violence caused by a spouse, common-law partner, or dating partner within an intimate relationship. It includes physical violence such as punching or kicking, sexual violence such as rape or the use of manipulation to pressure someone into sex, and emotional and verbal abuse, such as financial abuse and belittling, threatening, and controlling behavior

There is a close link between intimate partner violence (IPV) and HIV. IPV can increase risk of HIV by limiting a person's ability to negotiate safer sex and safer drug use and because the short- and long-term effects from IPV may lead people to engage in higher risk behaviors. IPV is common among people living with HIV, in part because IPV and HIV disproportionately affect some of the same populations. People living with HIV may experience more severe or more prolonged IPV than people who are HIV-negative and experiencing IPV. People living with HIV may be particularly vulnerable to IPV when they disclose their HIV status to their partner. Intimate partner violence can lead to poorer HIV care outcomes. HIV service providers can play a role in helping to prevent or respond to IPV. Likewise, those working with people who experience IPV can play a role in helping to prevent HIV and encouraging testing and treatment.

3. Drug abuse and HIV/AIDS

Direct risk of HIV exposure occurs in the form of contaminated needles and drug injection supplies. When needles are shared among drug users, traces of blood may be transferred between individuals as well. 10% of new HIV infections are related to drug abuse

In addition to increasing the risk of HIV exposure, drug abuse may negatively affect HIV treatment and maintenance. For instance, chronic alcohol use weakens the immune system.

Weakened immune system promotes replication of the virus and increases the occurrence of OIs (opportunistic infections)

Using drugs affects your brain, alters judgment and lowers inhibitions. When someone is high, he/she is likely to make poor decisions that put them at risk of contracting HIV such as

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- Having sex without a condom
- Failure to use the condom in the right way
- Having more sexual partners
- Sharing needles with infected persons

People who inject drugs can reduce the risk of contracting HIV by:

- Avoiding sharing needles
- Using new sterile needles every time they inject
- Dispose off the needles effectively after use
- Avoiding sex when high
- Using prep and pep to avoid infection

WHO STAGING OF HIV INFECTION

Following HIV infection there is initial rapid rise in viral load, which may be associated with flue like symptoms(primary hiv infection). A vigorous immune response occurs within weeks of infection, with production of both antibodies and cellular mediated response.

Thus antibodies to HIV can be detected within 2 week of infection. The immune response inhibits HIV replication with the results that the viral load declines and stabilizes within the first 6-12 months after infection. Thereafter the patient remains asymptomatic for an average 6-10 years in the majority of patients, before the ongoing immune destruction begin to manifest in symptomatic HIV related disease

At the time of initial infection with HIV, patient have large number of susceptible CD4+ T-lymphocytes and no HIV specific response. Thus, there is rapid HIV replication that causes rapid destruction of CD4+ T lymphocytes over the first weeks and months after infection. Through the induction of HIV specific immune response, there is stabilization of CD4 cells level and containment of viral replication, which is marked by a very slow decline of CD4+ T-cells over 6-10 years. Eventually in the absence of effective arvs CD4 cell level decline further culminating in

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

progressive immune deficiency accompanied by the development of HIV associated complication(e.g.OIS,malignancies)

While immunologic decline after primary infection usually occur slowly over many years, it is worth noting that many patients will not be diagnosed until late in the course of infection, when symptoms are present and CD4 count is low.

WHO Clinical staging of HIV/AIDS

Clinical stage1;

1. Asymptomatic

2. Persistent generalized lymphadenopathy (PGL)

Clinical stage 2;

1. Moderate unexplained weight loss (<10% of presumed or measured BW)

2. Minormucocutanousmanifestation (seborrheic dermatitis, popular pruritic eruption,fungal nail infection,recurrent oral ulcerations,angular cheilitis)

3. Herpes zoster

4. Recurrent URTI(sinusitis,tonsilitis,bronchitis,otitis media, pharyngitis)

Clinical stage 3;

1. Unexplained severe weight loss(over 10% of presumed or measured BW)

2. Unexplained diarrhea for more than one month

3. Unexplained persistent fever

4. Persistent oral candidiasis

5. Oralhairy leukoplakia

6. Pulmonary tuberculosis

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

7. Severe bacterial infection (e.g pneumonia, empyema,pyomyositis,bone and joint infection,menengitis,bacterimia.)

8. Unexplained anemia(below 8 g/dl)

Clinical stage 4

1. HIV wasting syndrome

2. Pneumocysticjiroveci pneumonia (pcp)

3. Recurrent severe bacteria pneumonia (2 episodes within 1 year)

4. Cryptococcal meningitis

5. Toxoplasmosis of the brain

6. Kaposi sarcoma (ks)

7. HIV encephalopathy

8. Extra pulmonary tuberculosis

Opportunistic infections

Also called OIs.

They are infections that occur more frequently and are more severe in people with weakened immune system esp. those with HIV. PLWHA, especially people not on ART, can get a lot of OIs

One of the best ways to live positively with HIV/ AIDS whether you are on ART or not, is to prevent opportunistic infections from happening in the first place or treating them right away if they do happen. The most common OIS are;

1. Tuberculosis

2. Malaria

3. Pneumonia or pcp

4. Meningitis5. Sexually transmitted infections

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

6. Diarrhea

7. Vomiting

8. Skin problems

9. Oral sores

More about TB is the most common OI among PLWHA. TB and HIV are like brother and sister- about 50% of PLWHA in Kenya also have TB and about 70% of people with TB also have HIV Because TB and HIV are so closely linked, services to prevent and treat both also need to be linked.

TB is spread through the air, especially from coughing, sneezing and being in close quarters with a person with TB. It's made worse when there is not good air circulation like if there are no open windows. It's very easy to spread TB from person to person, and PLWHA are especially likely to get it if they are in contact with a person with TB. Little children are at high risk for TB when they live with adult who have TB

Key Risk Factors Fuelling the Spread of HIV/ AIDS

It explains African traditional beliefs and practices which may be major factors in the spread of the virus causing AIDS. It gives myths on sexuality and links them to the rapid spread of HIV/AIDS. Also discusses socio-economic and cultural factors and also political factors influencing the spread of HIV/AIDS.

a .Myths /mysteries surrounding HIV/AIDS

People lack information about HIV/AIDS They think it is a curse from God, witch craft, originated from monkeys, lab mistakes, blame Americans who might have introduced it to African sex or controlling population

Some of the myths which need to be demystified include;

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Having multiple sex partners boosts men prestige and status among the peers(not true)

Abstinence causes male impotence(not true)

Sex with a virgin provides a cure for a man infected with HIV/AIDS

Teaching young people about sex encourages them to practice”(not true)

Ignorance about sex is a sign of innocence and purity while too much knowledge is a sign of immorality(information is power)

Knowing your status means early death(not true)

ARVS provide cure for AIDS (not true)

b.Socio-economic and cultural factors

Pre-marital sex-before marriage involve fornication and adultery Age at 1st sexual encounter-women aged between 14-25 majority infected Drug and substance abuse common among youth in institution of higher learning Lack of control over sexual urge thus going for risk sex

Inequality in making decisions In access and control over assets/resources at family community and national level Women obliged/compelled/required to have sex when she is aware that partner is infected Having sex besides the matrimonial spouse African traditional ceremonies-long duration, social gathering and festivities

Wife inheritance and window cleansing especially through sexual performance

Ear piercing

Circumcision Female genital mutilation(FGM)

Tattooing Teeth extraction Forced marriage poverty

Illiteracy

Unwillingness to know ones HIV status

Trial marriages

Sex expediency

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Sexual orientation

Resistance to condom use

Media and pornography

Tendency to copy and put to practice

Youth idle and engage in risk behavior

c.**Political factors**

Decreased allocation of resources and commitment to the implementation of policies that exist from external donors

Sustainability –mobilization of locally generated resources

Most of the intentioned policies which have been formulated, have not been fully implemented Corruption

MOST AT RISK POPULATIONS (MARPS)

Anyone can contract HIV, but there are certain groups of people that are more affected than others:

1. **Men who have sex withmen** (MSM). Gay or bisexual MSM are most severely affected population. This is because they have sex on the anus which is prone to tearing due to lack of lubrication thus increasing the risk of HIV transmission

In order to prevent transmission of HIV/AIDS there is need for them to use condoms and lubrications such as ky jelly every time they have sex.Also get to know their HIV status often.

2. **Injecting drug users (IDUs).** These are people who abuse drugs and mostly they share syringes and needles. Also by abusing drugs it make them vulnerable to having unprotected sex which put them at a higher risk of contracting HIV/AIDS.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

In order to protect themselves they should avoid sharing of syringes and needles and also use condoms when they engage in sexual activities.

3. **Track drivers.** They travel for long distance i.e from one town to another leaving their spouses behind .they get involved sexually with multiple partners which put them at a risk of getting infected with HIV/AIDS In order to avoid this they should know their HIV status often, use condoms and practice faithfulness.

4. **Prisoners**. They are most at risk because they practice homosexuality which is illegal in our country. Use of condoms in the prison is illegal which put them at a greater risk In order to reduce the risk of transmission there should be provision of condoms, ARVS and also HIVtesting and counselling.

5. **Fish forks/mongers**. Another most at risk group they leave their families to do fish business at the lakes and involve themselves with multiple sex partners. In order to prevent transmission there is need to do HIV testing regularly,use of condoms and faithfulness.

6. **Commercial sex workers both men and women.** They are at risk because of multiple sex partners. In order to prevent the risk of HIV transmission there is need to regularly know their HIV status, use of condoms and treatment of STI'S.

7.**Young people**. They are most at risk because of peer influence, hormonal changes, experimenting and many more they have multiple sex partners without using protection.

• In order to prevent the risk of transmission they should use condoms, know their HIV status, and avoid negative peer influence.

<u>Opportunistic infections</u>

Also called OIs. They are the infections that make PLWHA because the body immune system is weakened and it cannot fight back. When the HIV attacks the CD4 cells,the person has trouble fighting back when a virus enters their body.PLWHA,especially people not on ART,can get a lot of OIs

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

One of the best ways to live positively with HIV/ AIDS whether you are on ART or not, is to prevent opportunistic infections from happening in the first place or treating them right away if they do happen. The most common OIS are;

1. Tuberculosis

2. Malaria

3. Pneumonia or pcp

4. Meningitis5. Sexually transmitted infections

6. Diarrhea

7. Vomiting

8.Skin problems

9. Oral sores

More about TB is the most common OI amongPLWHA. TB and HIV are like brother and sister- about 50% of PLWHA in Kenya also have TB and about 70% of people with TB also have HIV Because TB and HIV are so closely linked, services to prevent and treat both also need to be linked.

TB is spread through the air, especially from coughing, sneezing and being in close quarters with a person with TB. It's made worse when there is not good air circulation like if there are no open windows. It's very easy to spread TB from person to person, and PLWHA are especially likely to get it if they are in contact with a person with TB. Little children are at high risk for TB when they live with adult who have TB

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

2. MANAGEMENT OF HIV/AIDS

INTRODUCTION;

Knowledge about HIV status remains a major limiting factor to accessing HIV prevention, care and treatment services, therefore at all HIV testing sites. VCT services should refer patients to HIV / comprehensive care clinic(ccc) as soon after diagnosis is made

1. PICT(provider initiated counselling and testing) should be carried out aggressively in and out patient service points (ANC,TB,STI,FP,MCH etc clinics) All patients with children under 15 years attending CCC should be encouraged to bring their children for hiv testing in additional to sexual partners
2. All community based organizations and families supporting orphans and vulnerable children should arrange to have those children whose parents may have died of HIV infection tested All people living with HIV ,whether on ART or not, should have access to a set of core interventions known to promote health,improve the quality of life ,prevent further HIV transmission ;and for some delay HIV disease progression and prevent mortality.
3. Counselling and psychosocial support; 1. All people living with HIV should be provided with counselling and psychosocial support intervention including individual and group counselling ,peer support groups ,family and couple counseling and support . Counseling and psychosocial support should focus on ;

- Mitigation of fear,anger,self-stigma and discrimination
- Alleviation of grief,bewilderment and stress among partners and family members
- Behaviour change in support of healthy living and prevention of further HIV transmission
- Disclosure and partner notification
- Family/partner counselling to identify family members who may need care and treatment
- Skills –building on how to live a healthy and productive life

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

• Identification and treatment of depression and substance abuse.Mental illness and substance and alcohol dependence are common conditions among PLWHIV.These conditions,besides affectingthe quality of life of patient can cause non adherence to prophylaxis and ART regimens as well as underminesafer sex practices and youth and adolescence issues

4. People living with HIV should be offered counselling and support to promote adherence to prevention interventions and treatment recommended for their care including ART 5.PWHIV who choose to be sexually active should be counseled on safer sexual practices to prevent HIV transmission to their sexual partners and avoid acquisition of STIs and HIV reinfection; and should be provided with condoms and appropriate contraceptive services and counseling

DRUGS USED IN HIV

HIGHLY ACTIVE ANTIRETROVIRAL THERAPY (HAART)

This is mainly a combination of ARVs. In early days of HIV management/treatment, monotherapy or single drugs were used to manage the disease. But this had a very poor outcome as the virus developed resistance to them.

The combination of the ARVs was found to be more successfully and with the majority of the patients being on HAART, the viral load drops at a level that is not detectable using most sensitive laboratory equipments.

Ideally the viral load should drop to 90% within two months of beginning the treatment.

There is a rapid rise in the CD4 cell count accompanied by improvement of the patient's health.

The opportunistic diseases also disappears

Requirements of HAART treatment

- The patient needs to adhere to therapy and take the necessary changes

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- The patient needs to take the drugs religiously and know that they are not free of side effects
- The patient needs to be on therapy forever as its cure
- The drugs have dietary restrictions and bad taste and the patient should understand this.
- Laboratory tests for both viral load and CD4 cell count determination need to be carried after every three months.
- Patients should start HAART treatment only when they can adequately access the drugs for a reasonable period of time.

Hiv medication can help lower your viral load, fight infections, and improve your quality of life. But even if you take them, you can still give HIV to others. They're not a cure for HIV.

The goals for these medicines are to :

Control the growth of the virus Improve how well your immune system works

Slow or stop symptoms Cause as few side effects as possible

• The FDA has approved more than two dozen antiretroviral drugs to treat HIV infection. They're often broken into six groups because they work in different ways. Doctors recommend taking a combination of at least two of them. This is called **antiretroviral therapy,** or ART.

• Your doctor will let you know specifically how you should take your medications. You need to follow the directions exactly, and you shouldn't miss even one dose. Or you could develop drug-resistant strains of HIV, and your medication may stop working.

• Some other medicines and supplements don't mix well with HIV drugs, so make sure you tell your doctor about everything you're taking.

TREATMENT OF HIV/AIDS

Antiretroviral drugs are drugs that stop HIV virus from multiplying inside the human cell. They are not cure. There are the classes of drugs which have been found effective in inhibiting and

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

surpassing the replication of the HIV virus. These are the drugs that have led to the reduction of HIV virus circulating in the blood. They improve the patient's immunological responses leading to a reversal of symptoms of the opportunistic diseases. They have led to a significant reduction of diseases due to HIV. They have improved the quality of life for those infected with the HIV virus.

Advantages of ARVs

1. They restore immune function and slows the decline of the immune system.
2. It prolongs life and improves the quality of life.
3. It reduces the risk of illness and hospitalization
4. It improves the symptoms of HIV opportunistic infections
5. They improve health and strength.

Disadvantages of ARVs

1. They are not cure and may raise false hope
2. They are taken for the remaining part of a patient's life.
3. At least three drugs have to be taken together to be effective.
4. Most of the regimens have complicated schedules
5. The drugs have side effects
6. If resistance develops, the drugs no longer work effectively
7. Most of these drugs are very expensive

CLASSIFICATION OF ARV DRUGS

Currently ,there are 5 classes of drugs active against HIV.ARV agents act by interfering with important functions in the viral life-cycle.

1.Nucleoside/Nucleotide Reverse Transcriptase Inhibitors (NRTIs)They work by prematurely terminating DNA chain formation as the enzyme reverse transcriptase copies viral RNA into DNA.Examples are Abacavir, or ABC (Ziagen) • Didanosine, or ddl (Videx) • Emtricitabine, or FTC

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

(Emtriva) • Lamivudine, or 3TC (Epivir) • Stavudine, or d4T (Zerit) • Tenofovir, or TDF (Viread) • Zidovudine, or AZT or ZDV (Retrovir)

2.Non-nucleoside Reverse Transcriptase Inhibitors (NNRTIs) Inhibit the formation of viral DNA from viral RNA by tightly binding to the reverse trascriptaseenzyme.Example are • Delavirdine (Rescriptor) • Efavirenz (Sustiva) • Etravirine (Intelence) • Nevirapine (Viramune) • Rilpivirine (Edurant)

3.Protease Inhibitors (PIs) They bind to the viral protease enzyme and block the formation of viral proteins. • Atazanavir, or ATV (Reyataz) • Darunavir, or DRV (Prezista) • Fosamprenavir, or FPV (Lexiva) • Indinavir, or IDV (Crixivan) • Lopinavir + ritonavir, or LPV/r (Kaletra) • Nelfinavir, or NFV (Viracept) • Ritonavir, or RTV (Norvir) • Saquinavir, or SQV (Invirase) • Tipranavir, or TPV (Aptivus)

• Cobicistat (Tybost) is a drug that helps atazanavir and darunavir work better, but it can cause life-threatening kidney problems if you take it with certain other medicines. • Atazanavir + cobicistat, or ATV/COBI (Evotaz) • Darunavir + cobicistat, or DRV/COBI (Prezcobix)

4.Entryinhibitors; they prevent entry of the virus into the host cell i.e CD4.Example include enfuvirtide CCR5 antagonist Maraviroc

5.Integrase strand transfer inhibitors(INSTI);they block the integrase enzyme which incorporates/integrates pro viral DNA into the host cell DNA.Exampleare;Raltegravir, Elvitegravir,Dolutegravir

SIDE EFFECTS OF ARVS

• HIV medications help many people lead longer, healthier lives. However, AIDS and HIV drug side effects are also common. These side effects range from mild to life threatening. Here is an overview of some of the common and more severe HIV drug side effects.

<u>COUNSELLING</u>

Counselling has been defined as a process of helping a person/people learn how to solve certain interpersonal, emotional and decisional problems. Counselling, in relation to HIV and

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

AIDS is a confidential dialogue between a person and a care provider aimed at enabling the person cope with stress and make informed personal decisions relating to HIV and AIDS (WHO 1994).

MAJOR GOALS OF HIV/AIDS COUNSELLING

The major goals of HIV/AIDS counselling are as follow:

a) Preventive: Providing counselling service and information to help prevent and mitigate the continued spread of HIV by providing information about risk behaviours that leave people vulnerable to contracting HIV infection as well as helping individuals to develop the required skills for behaviour change.

b) Supportive: Providing counselling services to help support people that are infected or affected by HIV. The support includes emotional, social and psychological help given to people who are infected by HIV and those that are affected by the virus.

c) Rehabilitative: Ensuring that clients have access to all the health services available by providing adequate referrals for treatment, care and support services.

PRETEST COUNSELLING

Counseling before an HIV test helps a person to understand:

- ✓ The meaning of HIV infection
- ✓ The meaning of the results- negative or positive
- ✓ Helps one to be prepared for the outcome

Pretest counseling should include;

- Information about the test
- Information about HIV infection and aids
- Why the person is being tested
- What it will mean if the test is positive/negative
- Discussion about confidentiality
- Education about HIV prevention and how to reduce risks
- Decision about whether or not to have test done.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- It will include the meanings of HIV tests including window period
- It includes reasons why the client is requesting VCT

POST TEST COUNSELLING

This one is a dialogue between a client and a counselor aimed at explaining the HIV test result, provides appropriate info, support and referral and encouraging risk reduction behavior

The client is helped to understand the test results and he is reassured and given hope

The counseling after HIV test can help a person to;

- Clarify information
- Understand the test
- Express t heir feelings and reactions
- It helps a person to change behavior that may place others at risk.

The diversified diet

This is part of the food that the body uses to produce energy, for movement, building or repairing body tissues, and protect the body from infections. Every meal should contain food from each of the following groups:

1. Energy giving foods- e.g. rice, maize, millet cassava, yams, sweet potatoes, etc. they should not be processed to give the most benefit. They provide the body with the main part of the meal.
2. Body building foods and proteins- e.g. chicken, fish, eggs, dairy products, beans, peas, nuts, etc. they contain proteins and minerals and some vitamins.
3. Vitamins- e.g. fruits, vegetables. They protect the body from infections
4. Fats-e.g. vegetables, nuts. They give energy. They should be used sparingly as they may cause diarrhea.
5. Water- it's a source of life. It's taken when clean and sterile.

Indicators of poor health

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Lack of appetite

Indigestion

Fever

Weight loss

HOME BASED CARE

This is the care of the patient by the HIV that is outside the hospital to the patient home through the family participation and community involvement using available resources and in collaboration with the health facility.

Hbc has been noted to be the best way to ensure that quality care can be continuously provided outside the health unit level.

Components of HBC

Clinical care- this includes early diagnosis and treatment and planning for follow up care for hiv related illnesses.

Nursing care- this includes care to promote and mentain good health, hygiene,nutrition and ultimately to assist terminal patient to a peaceful and dignified death.

Counseling and psycho-spiritual support: this includes stress and anxiety reduction, promoting positive living and helping individuals make informed decisions on hiv testing, planning for future and behavioural changes and involving sexual partners in such decisions.

Social support- this includes information and referral to support groups, welfare services and legal advice for individuals and families including some family membes and where possible provision of material assistance.

Who should provide HBC and where?

i. Only when patient is not very sick the patient can provide for himself

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

ii. Family members-relatives, friends and community
iii. Nurses and social workers
iv. Spiritual leaders
v. Hiv trainers

Advantages of home based care

- Cheaper compared to hospital
- Home environment allows patients to continue participating in family matters
- A patient who is nursed in a familiar environment usually suffers less stress and anxiety. This makes the illness more bearable.
- Being in constant contact with family members, friends and relatives help the patient accept the disease.
- Patients rest at home with family members and community as a whole and this promotes a sense of belonging.
- Good basic care can be given to the patients at home than in hospital.
- Information and education on hiv/aids will help family members to understand the patient's condition
- Training in home based care creates awareness on hiv/aids. It dilutes myths and beliefs about it and enhances prevention
- It reduces costs and encourages community participation in care of patients
- Discrimination and stigma is dealt with

Qualities of a good HBC giver

- Good communication
- Courage
- Friendly
- Patient
- Caring
- Humble
- Loving

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- Understanding
- Empathetic

NUTRITION AND HIV/AIDS

Eating well is an important part of keeping the body well and managing it hence the best intervention. People living with HIV have reduced appetite and this is due to infections, depression, side effects of medication, poor nutrient absorption

IMPORTANCE OF GOOD NUTRITION FOR PLWHIV

- Increases resistance to infections
- Improves energy
- Boosts the immune system
- Replaces lost micronutrients
- Preserves muscle mass
- Delays the rate of progression of HIV

It provides the body with all essential nutrients required for the body

The balanced diet

This is part of the food that the body uses to produce energy, for movement, building or repairing body tissues, and protect the body from infections. Every meal should contain food from each of the following groups:

- Energy giving foods- e.g. rice, maize, millet cassava, yams, sweet potatoes, etc. they should not be processed to give the most benefit. They provide the body with the main part of the meal.
- Body building foods and proteins- e.g. chicken, fish, eggs, dairy products, beans, peas, nuts, etc. they contain proteins and minerals and some vitamins.
- Vitamins- e.g. fruits, vegetables. They protect the body from infections

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- Fats-e.g. vegetables, nuts. They give energy. They should be used sparingly as they may cause diarrhea.
- Water- it's a source of life. It's taken when clean and sterile.

THE LINK BETWEEN FOOD, NUTRITION AND HIV/AIDS

Nutrition and HIV/AIDS are strongly interrelated. Malnutrition can both contribute to and result from the progression of HIV. This relationship between malnutrition and HIV/AIDS creates a vicious cycle: HIV weakens the immune system, which in turn leads to more infections (See Figure 10 below). Infections increase energy needs and at the same time cause anorexia, as indicated in Table 2. Heightened infections (in number and severity) lead to loss of appetite, resulting in inadequate food intake, and eventually malnutrition. Malnourished persons are at greater risk of infections, creating more vulnerability to HIV, and so the cycle continues.

Poor Nutrition

Increased vulnerability to infections, poor health, earlier and faster progression to the end stage of AIDS

Poor ability to fight HIV and Other infections HIV increased nutritional needs reduced food intake and increased loss of nutrients Poor nutrition among PLHIV may result from a number of causes such as:

1. Increased nutrient needs
2. Diarrhea which causes poor absorption resulting in poor utilization of ingested foods and nutrients
3. Reduced food intake due to:

- Reduced appetite as a result of infection
- Mouth infection including Candida which makes it painful to eat or suckle
- Depression
- Side effects of drugs
- Nausea
- Vomiting

An effective nutritional care and support program will improve the quality of life of people living with HIV/AIDS by:

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- Maintaining body weight and strength
- Replenishing lost vitamins and minerals
- Improving the function of the immune system and the body's ability to fight infections
- Prolong the period from infection to the development of the AIDS disease
- Improving response to treatment; reducing time and money spent on health care
- Keeping HIV-infected people active, allowing them to take care of themselves, their family and children; and
- Keeping HIV-infected people productive, able to work, grow food and contribute to the income of their families

Nutrient requirements for PLHIV

In PLHIV there is increased nutrient needs due to:

- Increased resting energy equilibrium (REE)
- Viral load (body trying to cope with illness/inflammation)
- Opportunistic infections (OIs) that increase energy demand
- Nutrient malabsorption

5. PREVENTION OF HIV INFECTION

INTRODUCTION:

Non pharmacologic interventions (abc)

The so-called ABC of HIV Prevention was developed by governments and their partners as the key approaching the fight against HIV/AIDS. The ABC messages standing for Abstain from sex, Be faithful to one partner and use a Condom, have been accepted and applied globally and are the dominant prevention messages.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

The approach consists of mutually reinforcing messages encouraging behaviour change with regards to sexual activities and should not be seen as stand alone package in HIV prevention programming. The messages are targeting the overall population and particularly sexual active age groups, both adults and youth.

Abstain from sex:The core message in the A is to stay abstinent until marriage. Abstinence is the most effective means of protection against both pregnancy and HIV. With sex being the primary cause of HIV transmission , it is unlikely for a person to contract the disease if he or she does not have sex.

• However, A of prevention appears not to be widely practiced – even though young people have positive attitudes towards the idea itself. The almost constant teenage pregnancy rates of about 19-20% of 15-19 year old girls between 1998 and 2002 are also a clear indicator that young people indeed are having sex.

Be faithful to one partner: The message contained in B is to be faithful to a mutually faithful and uninfected partner and reduce the number of sexual partners if possible to one. This is a rather complex concept since it requires commitment, honest y and trust among partners, without those this prevention strategy does not work. It is difficult to know for sure if your partner is being faithful, and it is clear that many partners are not. This is also true for youth, who are in an experimental stage sexually and emotionally.

Condom use: Consistent and correct use of condoms during intercourse can greatly reduce the chances of acquiring or transmitting HIV. Consistent use means using a condom each act of intercourse Correct use means that you should use anew condom every time you have sexual intercourse

Never use the same condom twice Put a condom before any contact is made between the penis and any part of your partners body Leave a space at the tip Withdraw immediately after ejaculation Condoms is not totally safe because it has some weakness which includes; Bursting Leaking Slipping off inside the woman-which exposes both male and female to hiv virus

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

In addition, the application of the C message is challenged by social norms and gender relations making it difficult for girls and women to negotiate condom use. Also, insecurities about the effectiveness of condoms have begun to undermine the status of condoms as a tool for HIVprevention among young people. Having experienced that condoms do not provide 100% protection from HIV-prevention if not used correctly or taken off early, some people don't feel secure even when condoms are being used.

• Surveys undertaken in different parts suggest that there are instances of young girls having unprotected sex in exchange for much-needed material support to the girls and their families including money for school fees, transportation and groceries or for luxury goods such as cell phones, toiletries etc. Poverty and inequality contribute to the rising numbers of young girls and boys having unprotected sex with partners that are much older, also popularly known as sugar daddies or sugar mummies

• Considering that studies demonstrate a strong correlation between high incidence of HIV and the age difference between partners, the sugar daddy phenomenon places young girls at high risk with regards to HIV. Older men are more likely to be infected with HIV than younger men, posing a greater risk of transmission to their younger partners.

• Giving 'gifts' or money to the girls may also make men feel as being in power and having the right to control the terms of the relationship. Also for women who find themselves in an abusive relationship or who are economically or otherwise dependent on a partner it is often difficult to negotiate sex and condom use. As a consequence,they are at high risk of contracting HIV/AIDS.

• Based on these communication messages and as part of the overall ABC approach, the Government together with its partners and young people concerned developed and adapted a variety of programmes in education, awareness-raising and life skills training. These programmes are going hand in hand with the ABC messages and are all geared towards encouraging behaviour change with regards to sexual activities aiming to stop the HIV/AIDS epidemic.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

• D – Delaying: Youth who begin sexual activity early appear more likely to have sex with high-risk partners or multiple partners and are less likely to use condoms. Thus early sexual debut can place adolescents at increased risk of unintended pregnancy, HIV, and other sexually transmitted infections. Positive relationships with parents, teachers, and spiritual beliefs decrease the likelihood of early sex, while engaging in other hazardous behaviours and having friends who are sexually active can be considered as risk factors.

• Therefore, encouraging youth to start engaging in sexual activities only when getting married or being mature enough to take responsible decisions need to go hand in hand with revisiting the important function parents, teachers and influential community members have as role models. Delaying is an important, practicable prevention message reflecting and responding to the realities of many youth and therefore needs to be added to the so-called ABC of HIV prevention.

2.**Prevention of mother to child transmission(PMTCT**)

Having a HIV test done before getting pregnant and during pregnancy Taking ARVS during pregnancy and delivery and breastfeeding and for life. Having a caesarean section instead of normal delivery i.e to avoid hurting the baby Avoiding episiotomy during derivery Giving the baby arvs after derivery

3.Prevention of transmission through blood and other blood product Screening all donated blood/all donated blood must be screened Careful handling of blood and blood fluids

Other methods of prevention are; Avoidance of sharp injuries-neddles,knives,clips, sharp objects in hospital working situation. All used sharp objects e.gneddles and blades should be disposed in the right way Never pick a sharp object without looking/ careful handling of sharp objects

Use of protective gear such as gloves(heavy duty gloves Avoid skin/mucous membrane contamination Equipment should be thoroughly and properly sterilized Avoid sharing nedddles and syringes Use of PEP taken after exposure to HIV Use of Prep taken before exposure by HIV

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

NEGATIVE people. Vmmc(voluntary medical male circumcision) Sex without penetration e.g masturbation ,massage etc

Voluntary medical male circumcision (VMMC)

Key facts Medical male circumcision reduces the risk of femaleto-male sexual transmission of HIV by approximately 60%. Since 2007, WHO and UNAIDS have recommended voluntary medical male circumcision as an additional important strategy for HIV prevention, particularly in settings with high HIV prevalence and low levels of male circumcision, where the public health benefits will be maximized. Fourteen countries in eastern and southern Africa with this profile have initiated programmes to expand male circumcision.

• Medical male circumcision offers excellent value for money in such settings. It saves costs by averting new HIV infections and reducing the number of people needing HIV treatment and care. A one-time intervention, medical male circumcision provides men life-long partial protection against HIV as well as other sexually transmitted infections. It should always be considered as part of a comprehensive HIV prevention package of services and be used in conjunction with other methods of prevention, such as female and male condoms.

Overview Male circumcision is surgical removal of the foreskin - the retractable fold of tissue that covers the head of the penis. The inner aspect of the foreskin is highly susceptible to HIV infections. Trained health professionals can safely remove the foreskin of infants, adolescents and adults (medical male circumcision).

Compelling evidence for recommendations In 2007, WHO and UNAIDS issued recommendations on medical male circumcision as an additional HIV prevention strategy based on strong and consistent scientific evidence. Three randomized controlled trials undertaken in Kisumu, Kenya, Rakai District, Uganda, and Orange Farm, South Africa have shown that medical male circumcision reduces the risk of sexual transmission of HIV from women to men by approximately 60%.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

• The most recent data from Uganda show that in the five years since the Uganda trial was completed, high effectiveness has been maintained among the men who were circumcised, with a 73% protective effect against HIV infection.

What are the benefits of Voluntary Medical Male Circumcision? Medical male circumcision is a one-time health intervention that provides men with lifelong partial protectionn against HIV. Apart from reducing HIV infection, male circumcision decreases the risk of urinary tract infections; reduces risk of sexually transmitted diseases in men; protects against cancer of the penis and prevents inflammation of the glands and foreskin altogether.

• Moreover, Male Circumcision also benefits women: improving hygiene, reducing STI's, and reducing the risk of cervical cancer (Human Papilloma Virus – HPV). Men's health is as much about women's health when it comes to sexually transmitted diseases. • Most recently, results from a 5-year follow up study in Uganda showed that men who chose surgical VMMC were 70% less likely to be infected with HIV than men who were not circumcised. Similar results were reported from extensive follow-up studies in Kenya and South Africa.

Five Facts about Voluntary Medical Male Circumcision (VMMC) 1. A circumcised man is up to 60% less likely to get infected with HIV. 2. It does not affect the size of your penis. 3. There is some pain, but not much. 4. After VMMC, you can do most things after just 2 or 3 days. 5. There is no proven effect on sexual pleasure.

How male circumcision protects you against HIV: Before circumcision: The inside of the foreskin is soft and moist and is more likely to get a tiny tear or sore that allows HIV to enter the body more easily. The foreskin itself contains many 'target cells' that allow HIV to enter the body easily.

FREQUENTLY ASKED QUESTIONS ON VMMC 1. What is Voluntary Medical Male Circumcision (VMMC)?VMMC is the removal of a man's foreskin. It is a widely used, simple and safe procedure. Men can be circumcised at any age. While many men in Kenya have already undergone the procedure, only 10% of Luo men in Nyanza are circumcised today. 2. Why should men get circumcised?VMMC can provide protection against HIV infection. A circumcised man is up to 60% less likely to get infected with HIV than an uncircumcised man.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

3. What are other benefits of VMMC? VMMC reduces the man's chances of getting other infections like syphilis and herpes. VMMC also gives some protection against penile cancer in men and cervical cancer in women.

4. How does being circumcised protect men from HIV? The inside of the foreskin is soft and moist and is more likely to get a tiny tear or sore that allows HIV to enter the body more easily. The foreskin itself contains many 'target cells' that allow HIV to enter the body easily. After circumcision, the skin on the head of the penis becomes thicker and is less likely to tear

5. Why is it important to know your HIV status prior to VMMC?Someone considering VMMC should know their HIV status in order to get the health benefits that VMMC provides. Clients who decline the test but still want to get circumcised must be respected for their decision. VMMC is not recommended for HIV positive men.

6. How is VMMC done? The procedure takes a short time. An injection is given at the base of the penis to make it go numb, so that no pain is felt while the foreskin is cut off. The wound is then stitched, cleaned and bandaged

7. What happens after the operation? The man rests a short while, and then goes home. After 3 days the bandage is removed. He cannot have sex for the next 6 weeks. For the next 7 days, it is important to keep the wound clean, and avoid heavy exercise.

• After 7 days, the man needs to return to the health facility for a check-up. If men experience any complications after surgery (like prolonged pain or bleeding), they should seek assistance at the clinic

8. Does VMMC affect sexual intercourse?Yes, because men cannot have sexual intercourse for 6 weeks after VMMC. The wound needs enough time to heal properly. VMMC does not affect sexual performance. 9. Why should I talk to my partner about VMMC?VMMC has many benefits for both partners and both need to understand the procedure. It is best to openly discuss why you want VMMC. You can also visit a counselor or health worker together to discuss the matter

P .E.P(post exposure prophylaxis)

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

• Prophylaxis is a treatment that helps you ward off an infection from bacteria, fungus, a virus, or a parasite. Post-exposure prophylaxis, or PEP, means you take it after you may have come into contact with one of those bugs. • If you've been exposed to HIV (the virus that causes AIDS), PEP is a course of two or three drugs that will make you less likely to be infected.

• With the right treatment, there's an 80% chance the virus will be stopped. But you have to take the full course of drugs, and not everyone follows through. Only 57% of people who start the treatment typically finish it. This may be because you have to take the medication for 28 days, and it can cause side effects. It also can be expensive.

Who Might Need PEP? PEP may help: People who think they might have been exposed to HIV during sex Drug users who have recently shared needles or other related items Health-care workers who think they've been exposed to HIV on the job PEP is for emergency situations only. It should not be used as a substitute for safe sex or new sterile needles.

If you're frequently exposed to HIV -- through your partner or shared needles, for example - talk with your doctor about pre-exposure prophylaxis (PrEP) instead. PEP uses higher doses of more drugs than PrEP . Timing You have to start PEP within 72 hours of possible exposure. After that, the treatment won't work. If you think you may have been exposed to HIV, get medical attention as soon as possible.

Pep should be discouraged after 72 hrs of exposure as there is no benefit(ensure early referral to nearest centre offering PEP if there are no local services. Pre-exsisting medical condition and any current medications being used by an exposed individual. Choice of an efficacious simplified regimen preferably in a fixed dose combination whenever possible to increase adherence by reducing number of pills and frequency of dosing. Provide on going counselling due to adherence and adverse reaction

Note: pep may be considered following RTA where there has been exposure to other peoples blood ; among police and prison staff who may be injured in the course of their work and in discordant couples following condom accident.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Risk assessment following exposure to various body fluids; low risk-type of exposure ;intact skin-source HIV negative ; material; saliva, tears,sweat, faeces, urine ,sputum and vomitus High risk;type of exposure; mucus membrane/non intact skin, percutaneous injury, material; semen, vaginalsecretions, synovial, pleural, pericardial, peritoneal, amniotic fluids, blood and bloody bodily fluids, viral culture in laboratory.

Of particular high risk are deep injuries,those involving hollow neddles with visible blood and those involving patients with high viral loads (recent HIV infection,late stage HIV disease. Note;the transmission of HIV is probably significantly higher in rape because of trauma forceful penetration. Other factors that increase transmission risk include disease status of rapists(risk increase with viral load) and presence of STI'S in the source or the person assulted.In high prevalence population rapists should be assumed to be HIV-positive unless proven otherwise.

Additional service for sexual assault; Emergency contraceptives for a child bearing nonpregnant women STI prophylaxis rx to all Adult non pregnant Po Doxycline 100mg bd x7 days = ponorfloxacin 800mg stat Adult pregnant Iminjspectinomycin 2gm stat +po erythromycin 500mg qid x 7 days

Children ; Po amoxyllin 15mg/kg/dose TDS X7 days + Po erythromycin 10 mg/kg/dose qid x7 days

Post-exposure prophylaxis

In occupational exposure; Immediate care to exposure site;encourage bleeding from the site but do not scrub or cut the site,washing it with soap and running water Determine risk associated with exposure;evaluate the source and exposed person Asses the potential risk of infection Both the source and exposed person need to be counselled for hiv testing

A known source should be tested for hiv;if the source person is not willing to be tested he/she should not be coerced into having the test Discarded sharps/neddles should not be tested The exposed person should not receive ARVS drugs without being tested.However where immediate testing is not feasible,rx should be delayed since hiv testing can be carried out the following day or soon thereafter

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Hiv test should be done at baseline,at 3 months and 6 months for a person exposed. Offer PEP;Used FDC if available,to reduce pill burden and increase adherence. TDF/AZT +3TC +LPV/r ARV prophylaxis in sexual assault is different; TDF +3TC +LPV/r(adult only) Or AZT +3TC +LPV/r(for adult and children)

How It Works The idea behind PEP is that the same drugs that treat HIV can help you fight the virus as it tries to infect you. That involves a combination of two or three drugs called anti-retrovirals. The process is called antiretroviral therapy, or ART. The medications help keep HIV from making copies of itself. You'll take them once or twice a day for 28 days:

• For adults, the World Health Organization (WHO) recommends a drug called tenofovir, combined with either lamivudine, also called 3TC, or emtricitabine, commonly known as FTC. You might also be prescribed lopinavir, which is sometimes called LPV/r. • If a child under 10 needs PEP, the WHO recommends zidovudine, commonly known as AZT, and 3TC. If a third drug is needed, it recommends LPV/r.

• If you're on PEP, use condoms if you have sex to lower the chances that you'll be exposed to HIV again or that you'll pass the virus to others if you are infected.

PMTCT (emtc)

Means prevention of mother to child transmission of HIV. emtc means elimination of mother to child transmission of HIV. More than 90% of paediatric HIV infections are as a result of MTCT In absence of any intervention,the risk of MTCT is 15-30% in a non-breastfeeding population and 20-40% among women who practice prolonged breastfeeding without ARVS prophylaxis.The greatest risk of transmission is during labour and delivery due to the increased exposure of the newborn to HIV contaminated blood and body fluids.

Higher viral load(>=1000 copies/ml),lower CD4 counts(,=350 cells/mm3),rupture of amniotic membranes for more than 4 hours before delivery and prolonged breastfeeding all increases the risk of MTCT. Prematurity ,low birth weight,mastitis and genital tract infection are some additional factors that may further increase the risk

The four prongs of comprehensive MTCT

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

1. Primary prevention

2. Prevention of unintended pregnancies

3. Chronic care and support for HIV+women, their infant and families

4. Interventions to reduce HIV transmission to infant during pregnancy, labour and delivery

Recommended care of a HIV positive woman

1.All HIV infected women who desire pregnancy should receive preconception care to optimize their health status prior to pregnancy

2.All pregnant women should be encouraged to start attending antenatal care (ANC) as soon as they know that they are pregnant, preferably in the 1st trimester

3.All pregnant women should be offered HIV counselling and testing during their 1st ANC visit.Those negative should be re-tested after 3 months

4.All those pregnant who are not tested or opt out or decline HIV testing during the 1st ANC should be offered continued counselling and testing in subsequent visit

5.Health care workers should offer counselling and testing to all sexual partners of the ANC clients and children of all HIV infected.HIV +partner in doscordant relationship should be evaluated for ARVS eligibility.

6.If a HIV + at the time of enrolment into ANC or becomes pregnant while in care;a full baseline assessment should be performed including clinical psychosocial and laboratory assessment including Cd4 cell count and eligibility for ART initiation is determined

7.Whereavailable,all HIV –infected pregnant women should be screaned for hepatitis B virus infection and managed accordingly

8.Labour and derivery should be follow optimal obstetrics management guidelines

9.All HIV + pregnant women should be started on ART.By enrolling them in care.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Prefered 1st line ART regimen in pregnant women is; TDF+3TC+EFV Infant nevirapine prophylaxis; HIV exposed infants of women on ART should receive 6 week of daily nevirapine irrespective of breastfeeding practices

Infant who are breastfeeding whose mothers are not on ART should receive daily nevirapine until one week after complete ceasation of breastfeeding Infant who are not breastfeeding should receive 6 weeks of daily nevirapine. NB;AZT 15 mg/kg twice daily for 6 weeks is an alternative for infants who are not breast fed or whose mothers are on ART/T riple prophylaxis

3TC is an alternative for infants with severe NVP toxicity(grade 3 or 4) or if baby is on TB treatment with rimfapicin containing regimen. Dosage;2-4 mg/kg twice daily

Prevention with positives;

People living with HIV should be support and encouraged to disclose their HIV status to those who need to know ;particularly sexual partner. In Kenya,upto 7-11% of couples in stable long –term parnership are sero discordant HIV negative partners in a sero –discordant relationship are at high risk of HIV infection Disclosure of HIV status encourages couple counselling and testing the discussion of reproductive health issues such as desired to have children, better couple communication and condom use.

Accumlating evidence has shown that treatment of HIV positive partners in a sero –discordant relationship markedly reduces the risk of HIV transmission to the HIV negative partner. Sero –concordant couples also benefit from disclosure;thus facilitating couple counseling and testing ;provision of condoms to avoid STIs and unintended pregnancies and discussion of reproductive health matters.

BEHAVIOR CHANGE COMMUNICATION

Behavior change communication (BCC) is an interactive process with communities to develop tailored messages and approaches using a variety of communication channels to develop

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

positive behaviors; promote and sustain individual, community and societal behavior change; and maintain appropriate behaviors.

In the context of the AIDS epidemic, BCC is an essential part of a comprehensive program that includes both services (medical, social, psychological and spiritual) and commodities (e.g., condoms, needles and syringes). Before individuals and communities can reduce their level of risk or change their behaviors, they must first understand basic facts about HIV and AIDS, adopt key attitudes, learn a set of skills and be given access to appropriate products and services. They must also perceive their environment as supporting behavior change and the maintenance of safe behaviors, as well as supportive of seeking appropriate treatment for prevention, care and support

The role of behavior change communication

Increase knowledge. BCC can ensure that people are given the basic facts about HIV and AIDS in a language or visual medium (or any other medium that they can understand and relate to).

Stimulate community dialogue. BCC can encourage community and national discussions on the basic facts of HIV/AIDS and the underlying factors that contribute to the epidemic, such as risk behaviors and risk settings, environments and cultural practices related to sex and sexuality, and marginalized practices (such as drug use) that create these conditions. It can also stimulate discussion of healthcare seeking behaviors for prevention, care and support

Promote essential attitude change. BCC can lead to appropriate attitudinal changes about, for example, perceived personal risk of HIV infection, belief in the right to and responsibility for safe practices and health supporting services, compassionate and non-judgmental provision of services, greater open-mindedness concerning gender roles and increasing the basic rights of those vulnerable to and affected by HIV and AIDS. Y

Reduce stigma and discrimination. Communication about HIV prevention and AIDS mitigation should address stigma and discrimination and attempt to influence social responses to them.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Create a demand for information and services. BCC can spur individuals and communities to demand information on HIV/AIDS and appropriate services.

Advocate. BCC can lead policymakers and opinion leaders toward effective approaches to the epidemic.

Promote services for prevention, care and support. BCC can promote services for STIs, intravenous drug users (IDUs), orphans and vulnerable children (OVCs); voluntary counseling and testing (VCT) for mother-to-child transmission (MTCT); support groups for PLHA; clinical care for opportunistic infections; and social and economic support. BCC is also an integral component of these services.

Improve skills and sense of self-efficacy. BCC programs can focus on teaching or reinforcing new skills and behaviors, such as condom use, negotiating safer sex and safe injecting practices. It can contribute to development of a sense of confidence in making and acting on decisions.

BCC goals:

- Increase perception of risk or change attitudes toward use of condoms
- Increase demand for services
- Create demand for information on HIV and AIDS
- Create demand for appropriate STI services
- Interest policymakers in investing in youth-friendly VCT services (services must be in place)
- Promote acceptance among communities of youth sexuality and the value of reproductive health services for youth (services must be in place)
- BCC goals are related to specific issues identified when assessing the situation, knowledge, attitudes and skills that may need to be changed to work toward behavior change and program goals.

Guiding Principles

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- BCC should be integrated with program goals from the start. BCC is an essential element of HIV prevention, care and support programs, providing critical linkages to other program components, including policy initiatives.
- Formative BCC assessments must be conducted to improve understanding of the needs of target populations, as well as of the barriers to and supports for behavior change that their members face (along with other populations, such as stakeholders, service providers and community).
- The target population should participate in all phases of BCC development and in much of implementation.
- Stakeholders need to be involved from the design stage.
- Having a variety of linked communication channels is more effective than relying on one specific one. Pre-testing is essential for developing effective BCC materials.
- Planning for monitoring and evaluation should be part of the design of any BCC program. BCC strategies should be positive and action-oriented.
- PLWHA should be involved in BCC planning and implementation.

Challenges of communication

Illiteracy

Lack of access to sources of information

Ignorance as people fail to read articles related to HIV

SEXUALLY TRANSMITTED DISEASES

Sexually transmitted diseases (STDs) — or sexually transmitted infections (STIs) — are generally acquired by sexual contact. The organisms (bacteria, viruses or parasites) that cause sexually transmitted diseases may pass from person to person in blood, semen, or vaginal and other

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

bodily fluids. Sometimes these infections can be transmitted non-sexually, such as from mother to infant during pregnancy or childbirth, or through blood transfusions or shared needles.

STDs don't always cause symptoms. It's possible to contract sexually transmitted diseases from people who seem perfectly healthy and may not even know they have an infection.

STDs are considered public health problem because

- They affect people's health
- High cost of treatment
- Can lead to premature death
- Easily transmitted through sex
- Some STDs do not show signs hence easily transmitted without being realized
- People do not seek for treatment due to fear/stigmatization

Symptoms

Sexually transmitted diseases (STDs) or sexually transmitted infections (STIs) can have a range of signs and symptoms, including no symptoms. That's why they may go unnoticed until complications occur or a partner is diagnosed. Signs and symptoms that might indicate an STI include:

- Sores or bumps on the genitals or in the oral or rectal area
- Painful or burning urination
- Discharge from the penis

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- Unusual or odd-smelling vaginal discharge
- Unusual vaginal bleeding
- Pain during sex
- Sore, swollen lymph nodes, particularly in the groin but sometimes more widespread
- Lower abdominal pain
- Fever

Signs and symptoms may appear a few days after exposure, or it may take years before you have any noticeable problems, depending on the organism

Causes

Sexually transmitted diseases (STDs) or sexually transmitted infections (STIs) can be caused by:

- Bacteria (gonorrhea, syphilis, chlamydia)
- Parasites (trichomoniasis)
- Viruses (human papillomavirus, genital herpes, HIV)

Sexual activity plays a role in spreading many other kinds of infections, although it's possible to be infected without sexual contact. Examples include the hepatitis A, B and C viruses, shigella, and Giardia intestinalis.

Risk factors

Anyone who is sexually active risks some degree of exposure to a sexually transmitted disease (STD) or a sexually transmitted infection (STI). Factors that may increase that risk include:

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- **Having unprotected sex.** Vaginal or anal penetration by an infected partner who isn't wearing a latex condom significantly increases the risk of getting an STI. Improper or inconsistent use of condoms can also increase your risk.

  Oral sex may be less risky, but infections can still be transmitted without a latex condom or a dental dam — a thin, square piece of rubber made with latex or silicone.

- **Having sexual contact with multiple partners.** The more people you have sexual contact with, the greater your risk. This is true for concurrent partners as well as monogamous consecutive relationships.
- **Having a history of STIs.** Having one STI makes it much easier for another STI to take hold.
- **Misuse of alcohol or use of recreational drugs.** Substance misuse can inhibit your judgment, making you more willing to participate in risky behaviors.
- **Injecting drugs.** Needle sharing spreads many serious infections, including HIV, hepatitis B and hepatitis C.
- **Being young.** Half the STIs occur in people between the ages of 15 and 24 because they are sexually active and want to experiment.

Transmission from mother to infant

Certain STIs — such as gonorrhea, chlamydia, HIV and syphilis — can be passed from an infected mother to her child during pregnancy or delivery. STIs in infants can cause serious problems or even death. All pregnant women should be screened for these infections and treated.

<u>Complications of STIs</u>

Because many people in the early stages of a sexually transmitted disease (STD) or sexually transmitted infection (STI) experience no symptoms, screening for STIs is important in preventing complications.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Possible complications include:

- Pelvic pain
- Pregnancy complications
- Pelvic inflammatory disease
- Infertility
- Certain cancers, such as HPV-associated cervical and rectal cancers

Prevention

There are several ways to avoid or reduce your risk of sexually transmitted diseases (STDs) or sexually transmitted infections (STIs).

- **Abstain.** The most effective way to avoid STIs is to not have (abstain from) sex.
- **Stay with one uninfected partner.** Another reliable way of avoiding STIs is to stay in a long-term mutually monogamous relationship in which both people have sex only with each other and neither partner is infected.
- **Wait and test.** Avoid vaginal and anal intercourse with new partners until you have both been tested for STIs. Oral sex is less risky, but use a latex condom or dental dam to prevent direct (skin-to-skin) contact between the oral and genital mucous membranes.
- **Get vaccinated.** Getting vaccinated early, before sexual exposure, is also effective in preventing certain types of STIs. Vaccines are available to prevent human papillomavirus (HPV), hepatitis A and hepatitis B.

  The hepatitis B vaccine is usually given to newborns, and the hepatitis A vaccine is recommended for 1-year-olds. Both vaccines are recommended for people who aren't already immune to these diseases and for those who are at increased risk of infection, such as men who have sex with men and IV drug users.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- **Use condoms consistently and correctly.** Use a new latex condom for each sex act, whether oral, vaginal or anal.
- **Don't drink alcohol excessively or use drugs.** If you're under the influence, you're more likely to take sexual risks.
- **Communicate.** Before any serious sexual contact, communicate with your partner about practicing safer sex. Be sure you specifically agree on what activities will and won't be OK.
- **Consider male circumcision.** There's evidence that male circumcision can help reduce a man's risk of acquiring HIV from a woman who is infected (heterosexual transmission) by as much as 60%. Male circumcision may also help prevent transmission of genital HPV and genital herpes.
- **Consider pre exposure prophylaxis (PrEP).** The Food and Drug Administration (FDA) has approved the use of the combination drugs emtricitabine plus tenofovir disoproxil fumarate (Truvada) and emtricitabine plus tenofovir alafenamide (Descovy) to reduce the risk of sexually transmitted HIV infection in people who are at very high risk.

1. **Chlamydia**

Chlamydia is a bacterial infection of your genital tract. Chlamydia may be difficult to detect because early-stage infections often cause few or no signs and symptoms. When they do occur, symptoms usually start one to three weeks after you've been exposed to chlamydia and may be mild and pass quickly.

Signs and symptoms may include:

- Painful urination
- Lower abdominal pain
- Vaginal discharge in women
- Discharge from the penis in men
- Pain during sexual intercourse in women
- Bleeding between periods in women

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- Testicular pain in men

2. **Trichomoniasis**

Trichomoniasis is a common STI caused by a microscopic, one-celled parasite called Trichomonasvaginalis. This organism spreads during sexual intercourse with someone who already has the infection.

The organism usually infects the urinary tract in men, but often causes no symptoms. Trichomoniasis typically infects the vagina in women. When trichomoniasis causes symptoms, they may appear within five to 28 days of exposure and range from mild irritation to severe inflammation.

Signs and symptoms may include:

- Clear, white, greenish or yellowish vaginal discharge
- Discharge from the penis
- Strong vaginal odor
- Vaginal itching or irritation
- Itching or irritation inside the penis
- Pain during sexual intercourse
- Painful urination

Genital herpes symptoms

Genital herpes is a highly contagious STI caused by a type of the herpes simplex virus (HSV) that enters your body through small breaks in your skin or mucous membranes. Most people with HSV never know they have it, because they have no signs or symptoms or the signs and symptoms are so mild they go unnoticed.

When signs and symptoms are noticeable, the first episode is generally the worst. Some people never have a second episode. Others, however, can have recurrent episodes for decades.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

When present, genital herpes signs and symptoms may include:

- Small red bumps, blisters (vesicles) or open sores (ulcers) in the genital and anal areas and areas nearby
- Pain or itching around the genital area, buttocks and inner thighs

Ulcers can make urination painful. You may also have pain and tenderness in your genital area until the infection clears. During an initial episode, you may have flu-like signs and symptoms, such as a headache, muscle aches and fever, as well as swollen lymph nodes in your groin.

In some cases, the infection can be active and contagious even when sores aren't present.

3. Human papillomavirus (HPV) infection and genital warts symptoms

HPV infection is one of the most common types of STIs. Some forms of HPV put women at high risk of cervical cancer. Other forms cause genital warts. HPV usually has no signs or symptoms. The signs and symptoms of genital warts include:

- Small, flesh-colored or gray swellings in your genital area
- Several warts close together that take on a cauliflower shape
- Itching or discomfort in your genital area
- Bleeding with intercourse

Often, however, genital warts cause no symptoms. Genital warts may be as small as 1 millimeter in diameter or may multiply into large clusters. Warts can also develop in the mouth or throat of a person who has had oral sex with an infected person.

4. Hepatitis

Hepatitis A, hepatitis B and hepatitis C are all contagious viral infections that affect your liver. Hepatitis B and C are the most serious of the three, but each can cause your liver to become inflamed.

Some people never develop signs or symptoms. But for those who do, signs and symptoms may occur several weeks after exposure and may include:

- Fatigue

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- Nausea and vomiting
- Abdominal pain or discomfort, especially in the area of your liver on your right side beneath your lower ribs
- Loss of appetite
- Fever
- Dark urine
- Muscle or joint pain
- Itching
- Yellowing of your skin and the whites of your eyes (jaundice)

Relationship between STIs and HIV

If you have been diagnosed with syphilis you should also test for HIV. Having an STI, including syphilis, can increase your risk of getting HIV. This is because having an STI, especially one that causes sores, makes it easier for HIV to get into your body and cause an infection.

People living with HIV can also be more likely to get syphilis. This is especially the case for people who aren't on treatment or who have a lower CD4 count. You're more vulnerable to infections, like syphilis, if your immune system is weaker. Syphilis may also progress more quickly in people living with HIV, so if you're living with HIV and think there's a chance you might have syphilis it's important to get tested and treated early on

People living with HIV can contract other sexually transmitted infection due to their weakened immune system. The STDs can therefore come in as opportunistic infections.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)